

		BYTE 0	BYTE 1	BYTE 2		BYTE 3	BYTE 4	BYTE 5	BYTE 6	BYTE 7	BYTE 8	BYTE 9	BYTE 10	BYTE 11	BYTE 12	BYTE 13	BYTE 14	BYTE 15	BYTE 16	BYTE 17	BYTE 18	BYTE 19	BYTE 20	BYTE 21	BYTE 22	BYTE 23
DEFAULT_RESPONSE	NAME	FRAME ID	ERROR VECTOR		MOTOR TEMPERATURE		MAIN ENCODER POSITION			MAIN ENCODER VELOCITY				MOTOR TORQUE				OUTPUT ENCODER POSITION				OUTPUT ENCODER VELOCITY				
	TYPE	uint8_t	uint16_t		uint8_t [°C]		float [rad]			float [rad/s]				float [Nm]				float [rad]				float [rad/s]				
	VALUE	0x0A	0x0000-0xFFFF		0 - 255		-			-				-				-				-				

Communication Definition						ID	RESERVED	Data																							
FRAME NAME	ID	Watchdog	DEFAULT RESPONSE	Contents	LENGTH	BYTE 0	BYTE 1	BYTE 2	BYTE 3	BYTE 4	BYTE 5	BYTE 6	BYTE 7	BYTE 8	BYTE 9	BYTE 10	BYTE 11	BYTE 12	BYTE 13	BYTE 14	BYTE 15	BYTE 16	BYTE 17	BYTE 18	BYTE 19	BYTE 20	BYTE 21	BYTE 22	BYTE 23	BYTE 24	BYTE 25
FLASH_LED	0x00	Yes	Yes	to flash the onboard MD80 LEDs	2	0x00	0x00																								
MOTOR_ENABLE	0x01	Yes	Yes	to turn on the drive and begin the operation. After enabling the actuator, it is in IDLE motion mode, unless other mode was entered before.	3	0x01	0x00	Enable																							
								0x01 enable 0x00 disable																							
CONTROL_SELECT	0x02	Yes	Yes	to select the current motion mode	3	0x02	0x00	Mode																							
								0 - IDLE (shorted to GND, the motor is braking) 1 - position PID 2 - velocity PID 3 - deprecated(Not Used)																							
ZERO_ENCODER	0x03	Yes	Yes	to set the current shaft position to be the new zero position	2	0x03	0x00																								
SET_MAX_CURRENT	0x04	Yes	Yes	Set max current frame is used to set the max current limit. This limit is imposed on q-axis current, thus it applies to phase currents, not source current.	6	0x04	0x00	Current																							
								Byte 0(float)	Byte 1(float)	Byte 2(float)	Byte 3(float)																				
GET_INFO	0x05	Yes	Yes	to get the default response from the actuator	2	0x05	0x00																								
SET_TORQUE_BANDWIDTH	0x06	No	Yes	to modify the torque bandwidth of the actuator. Please refer to the manual for detailed information on torque bandwidth	2	0x06	0x00	Bandwidth																							
								(uint16_t) - torque bandwidth in Hz. Valid range: [50-2500]																							
SET_POSITION_PID	0x10	Yes	Yes	to set custom position PID gains(temporary gains)	32	0x10	0x00	Kp	Ki				Kd				i windup				Max output				Target position						
								(float) - proportional gain				(float) - integral gain				(float) - derivative gain				(float) - integral windup limit				(float) - maximum output limit (maximum velocity)				(float) - target position (rad)			
SET_VELOCITY_PID	0x11	Yes	Yes	to set custom velocity PID gains(temporary gains)	32	0x11	0x00	Kp	Ki				Kd				i windup				Max output				Target velocity						
								(float) - proportional gain				(float) - integral gain				(float) - derivative gain				(float) - integral windup limit				(float) - maximum output limit (maximum torque)				(float) - target velocity (rad/s)			
SET_IMPEDANCE_PD	0x12	Yes	Yes	to set custom impedance PID gains(temporary gains)	32	0x12	0x00	Kp	Kd				Target position				Target velocity				Feedforward torque				Max output						
								(float) - proportional gain				(float) - derivative gain				(float) - Position target				(float) - Velocity target				(float) - torque to be added at the controller output				(float) - maximum output limit (maximum torque)			
RESTART	0x13	Yes	Yes	to reboot the actuator	2	0x13	0x00																								
SET_MOTION_TARGETS	0x14	Yes	Yes	to set target position, velocity, torque, max velocity and max torque. It works with all modes, however not all fields are used in all modes.	24	0x14	0x00	Target velocity				Target position				Target torque				Max torque				Max velocity							
								applicable to all control modes				applicable to all control modes				applicable only to impedance mode				applicable to all modes (limits the maximum torque)				applicable to velocity mode (limits max velocity)							
SET_CAN_CONFIG	0x20	No	Yes	to change the FDCAN bus parameters such as ID, baudrate, and communication watchdog period, and optionally the software CAN bus termination	12	0x20	0x00	new ID		new baudrate				new wdg period		termination															
								(uint16_t) - FDCAN bus actuator ID (can be in range [10-2000])		(uint32_t) - baudrate (FDCAN bus speed) Valid : [1000000,2500000, 5000000, 8000000]				watchdog period Valid [0-2500ms]		(uint8_t) - available in HW 2.0 1 to turn on, 0 to turn off															
SAVE_CONFIG	0x21	No	Yes	to save the parameters in non-volatile memory	2	0x21	0x00																								
WRITE_REGISTER	0x40	No	Yes		X (64 max)	0x40	0x00	reg ID		value	...																				
READ_REGISTER	0x41	No	No		X (64 regx)	0x41	0x00	reg ID		0x00	...																				